

ORF 524 – Final Exam

Matias D. Cattaneo

Department of Operations Research and Financial Engineering
Princeton University

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Notes:

- Duration of examination: *4 hours*.
Additional 20 minutes are given to upload your solutions to Canvas.
- All parts are equally weighted (5 points).
Please provide as much detail as possible in your answers.
Answers without proper justification will not receive (partial) credit.
- Please start each question on a separate page.
A single PDF file must be submitted to Canvas.
- Only two pages of personal notes are allowed during the examination.
Basic auxiliary information about standard distributions is provided at the end of the exam.
This is otherwise a closed-book and individual examination.
Do not consult textbooks, online materials, or other sources.
Do not consult with classmates, other people or in online forums.

1 Question 1: Probability Theory and Asymptotics (30 points)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a background probability space. Recall that $\rightarrow_{\mathbb{P}}$ denotes convergence in probability and $\rightsquigarrow$ denotes convergence in distribution, with all limits taken as $n \rightarrow \infty$. In addition, recall that $\perp$ denotes (statistical) independence.

1. Let X be a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$.

Show that $X \perp X$ if and only if X is (a.s.) constant (i.e., $\mathbb{P}[X = c] = 1$, for some $c \in \mathbb{R}$.)

2. Let (X, Y, Z) be random variables on $(\Omega, \mathcal{F}, \mathbb{P})$.

Show that $\mathbb{E}[X\mathbb{E}[Y|Z]] = \mathbb{E}[Y\mathbb{E}[X|Z]]$, whenever both integrals exist.

3. Let $(X_n : n \geq 1)$ be a sequence of random variables on $(\Omega, \mathcal{F}, \mathbb{P})$.

Show that $|X_n|\mathbb{1}(|X_n| \geq M) \rightarrow_{\mathbb{P}} 0$ for all $M > 0$ whenever $\mathbb{E}[|X_n|^{1+\delta}] \rightarrow 0$ for some $\delta > 0$.

4. Let $(X_i : 1 \leq i \leq n)$ be a random sample with $X_i \sim \mathbb{P}$ and $\mathbb{E}[|X_i|^4] < \infty$. Let $\mu = \mathbb{E}[X_i]$, $\sigma^2 = \mathbb{V}[X_i] = \mathbb{E}[(X_i - \mu)^2]$, and $\mathbb{C}[X_i, X_i^2] = \mathbb{E}[(X_i - \mathbb{E}[X_i])(X_i^2 - \mathbb{E}[X_i^2])]$.

(a) Show that

$$\sqrt{n} \left(\begin{bmatrix} \frac{1}{n} \sum_{i=1}^n X_i \\ \frac{1}{n} \sum_{i=1}^n X_i^2 \end{bmatrix} - \begin{bmatrix} \mu \\ \sigma^2 + \mu^2 \end{bmatrix} \right) \rightsquigarrow \mathbf{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \mathbb{V}[X_i] & \mathbb{C}[X_i, X_i^2] \\ \mathbb{C}[X_i, X_i^2] & \mathbb{V}[X_i^2] \end{bmatrix} \right).$$

(b) Show that

$$\sqrt{n} \left(\begin{bmatrix} \bar{X}_n \\ S_n^2 \end{bmatrix} - \begin{bmatrix} \mu \\ \sigma^2 \end{bmatrix} \right) \rightsquigarrow \mathbf{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \mathbb{E}[(X_i - \mu)^3] \\ \mathbb{E}[(X_i - \mu)^3] & \mathbb{E}[(X_i - \mu)^4] - \sigma^4 \end{bmatrix} \right).$$

- (c) Give necessary and sufficient conditions for $\bar{X}_n$ and S_n^2 to be asymptotically independent. Give an example where $\bar{X}_n$ and S_n^2 are independent for all $n \geq 1$.

2 Question 2: Mean Absolute Error Optimality (20 points)

This question studies best Mean Absolute Error (MAE) prediction. Let $(Y, X) \in \mathbb{R}^2$ be an integrable random vector with joint probability law $\mathbb{P}$. The integrability assumption is imposed only for simplicity, but it is not really needed, and notice that no interchanging of differentiability and integration will be used. Let $\mathbb{P}_Y$ be the marginal probability law of Y , and $\mathbb{P}_{Y|X}$ be the (regular) conditional probability law of $Y|X$.

1. Show that if $\mathbb{P}_Y[Y \leq a] \geq 1/2$ and $a \leq b$, then for all $a, b \in \mathbb{R}$,

$$\mathbb{E}[|Y - b|] - \mathbb{E}[|Y - a|] \geq (a - b) \cdot (1 - 2 \cdot \mathbb{P}_Y[Y \leq a]) \geq 0.$$

Hint: $|Y - b| - |Y - a| = 2 \cdot (b - Y) \cdot \mathbb{1}(a < Y < b) + (a - b) \cdot [\mathbb{1}(Y > a) - \mathbb{1}(Y \leq a)]$.

2. Let m be a median of $\mathbb{P}_Y$, that is, $\mathbb{P}_Y[Y \leq m] \geq 1/2$ and $\mathbb{P}_Y[Y \geq m] \geq 1/2$.

Show that

$$\mathbb{E}[|Y - a|] \leq \mathbb{E}[|Y - b|].$$

for all $a, b \in \mathbb{R}$ such that $m \leq a \leq b$ or $m \geq a \geq b$,

3. Let $m(X)$ be a median of $\mathbb{P}_{Y|X}$. Show that

$$m(X) \in \arg \min_{T \in \mathcal{T}} \mathbb{E}[|Y - T(X)|],$$

where $\mathcal{T} = \{T(\cdot) : \mathbb{E}[|T(X)|] < \infty\}$.

4. Suppose $(Y, X)' \sim \mathbf{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, with

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_Y \\ \mu_X \end{bmatrix} \quad \text{and} \quad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{YY} & \sigma_{YX} \\ \sigma_{XY} & \sigma_{XX} \end{bmatrix}$$

Find m and $m(X)$. Are they (a.s.) unique?

3 Question 3: Statistical Inference (50 points)

Suppose $X_1, X_2, \dots, X_n$ is an i.i.d. sequence from $X \sim \mathbb{P}_\theta$, where $X_i \in \{0, 1\}$ for all i and $\theta = p_0 = \mathbb{P}_{p_0}[X_i = 1] \in (0, 1)$. Recall that $\rightarrow_{\mathbb{P}}$ denotes convergence in probability and $\rightsquigarrow$ denotes convergence in distribution. All limits are taken as $n \rightarrow \infty$, unless explicitly noted otherwise.

1. Show that the (proportion) estimator

$$\hat{p}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

is both a method of moments and the maximum likelihood estimator of p_0 .

2. Show that $\hat{p}_n$ is unbiased, and that $\mathbb{V}[\hat{p}_n] = p_0(1 - p_0)/n$. Is $\hat{p}_n$ a UMVUE of p_0 ?
3. Providing sufficient conditions, propose an estimator $\hat{V}_n$ satisfying $\hat{V}_n = \mathbb{V}[\hat{p}_n] + O_{\mathbb{P}}(n^{-3/2})$.
4. Providing sufficient conditions, show that

$$T_n := \frac{\hat{p}_n - p_0}{\sqrt{\hat{V}_n}} \rightsquigarrow Z,$$

where $Z \sim \mathcal{N}(0, 1)$.

5. Show that $\mathbb{P}_{p_0}[p_0 \in \text{CI}_{1-\alpha}] \rightarrow 1 - \alpha$, where

$$\text{CI}_{1-\alpha} = \left[\hat{p}_n - \Phi^{-1}(1 - \alpha/2) \cdot \sqrt{\frac{\hat{p}_n(1 - \hat{p}_n)}{n}}, \hat{p}_n + \Phi^{-1}(1 - \alpha/2) \cdot \sqrt{\frac{\hat{p}_n(1 - \hat{p}_n)}{n}} \right]$$

with $\Phi(t) := \mathbb{P}[Z \leq t] = \int_{-\infty}^t (2\pi)^{-1/2} e^{-u^2/2} du$.

6. A quantity often used in categorical data analysis is the “log-odds”:

$$\zeta_0 = \log \left(\frac{p_0}{1 - p_0} \right).$$

Construct an asymptotically valid $100(1 - \alpha)\%$ confidence interval for ζ_0 , where $\alpha \in (0, 1)$.

7. The confidence intervals above are only asymptotically valid *pointwise* in $\theta = p_0 \in (0, 1)$. However, their confidence coefficients are zero! That is, for $\text{CI}_{1-\alpha}$ above, we have

$$\inf_{p_0 \in (0, 1)} \mathbb{P}_{p_0}[p_0 \in \text{CI}_{1-\alpha}] = 0.$$

Show this result using the following two steps.

(a) Show that, for each fixed n ,

$$\mathbb{P}_{p_0}[p_0 \in \text{CI}_{1-\alpha}] = 1 - \mathbb{P}_{p_0}[p_0 \notin \text{CI}_{1-\alpha}] \leq 1 - \mathbb{P}_{p_0}[X_1 = 0, X_2 = 0, \dots, X_n = 0].$$

(b) Conclude that the confidence coefficient of $\text{CI}_{1-\alpha}$ is zero in finite samples, and hence asymptotically.

8. To better understand the previous result, let $(B_n : n \geq 1)$ be a sequence of random variables with $B_n \sim \text{Binomial}(n, p)$, where $n \in \mathbb{N}$ and $p \in (0, 1)$.

(a) Show that

$$\frac{B_n - np}{\sqrt{np(1-p)}} \rightsquigarrow \text{N}(0, 1),$$

which gives the well-known Gaussian approximation to the Binomial distribution.

This approximation is valid for “not too small” p , in the sense that $np \rightarrow \infty$.

(b) Suppose now that p is “small” in the sense that $np \rightarrow \lambda \in (0, \infty)$. Note this forces $p \rightarrow 0$ as $n \rightarrow \infty$. Use this embedding to show that

$$B_n \rightsquigarrow \text{Poisson}(\lambda),$$

which corresponds to the “law of rare events” approximation.

Table 1.1. Discrete Distributions on $\mathcal{R}$

Uniform	p.d.f.	$1/m, x = a_1, \dots, a_m$
	m.g.f.	$\sum_{j=1}^m e^{a_j t}/m, t \in \mathcal{R}$
$DU(a_1, \dots, a_m)$	Expectation	$\sum_{j=1}^m a_j/m$
	Variance	$\sum_{j=1}^m (a_j - \bar{a})^2/m, \bar{a} = \sum_{j=1}^m a_j/m$
	Parameter	$a_i \in \mathcal{R}, m = 1, 2, \dots$
Binomial	p.d.f.	$\binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \dots, n$
	m.g.f.	$(pe^t + 1 - p)^n, t \in \mathcal{R}$
$Bi(p, n)$	Expectation	np
	Variance	$np(1-p)$
	Parameter	$p \in [0, 1], n = 1, 2, \dots$
Poisson	p.d.f.	$\theta^x e^{-\theta}/x!, x = 0, 1, 2, \dots$
	m.g.f.	$e^{\theta(e^t - 1)}, t \in \mathcal{R}$
$P(\theta)$	Expectation	θ
	Variance	θ
	Parameter	$\theta > 0$
Geometric	p.d.f.	$(1-p)^{x-1}p, x = 1, 2, \dots$
	m.g.f.	$pe^t/[1 - (1-p)e^t], t < -\log(1-p)$
$G(p)$	Expectation	$1/p$
	Variance	$(1-p)/p^2$
	Parameter	$p \in [0, 1]$
Hyper-geometric	p.d.f.	$\binom{n}{x} \binom{m}{r-x} / \binom{N}{r}$
		$x = 0, 1, \dots, \min\{r, n\}, r - x \leq m$
$HG(r, n, m)$	m.g.f.	No explicit form
	Expectation	rn/N
	Variance	$rn m(N-r)/[N^2(N-1)]$
	Parameter	$r, n, m = 1, 2, \dots, N = n + m$
Negative binomial	p.d.f.	$\binom{x-1}{r-1} p^r (1-p)^{x-r}, x = r, r+1, \dots$
	m.g.f.	$p^r e^{rt}/[1 - (1-p)e^t]^r, t < -\log(1-p)$
$NB(p, r)$	Expectation	r/p
	Variance	$r(1-p)/p^2$
	Parameter	$p \in [0, 1], r = 1, 2, \dots$
Log-distribution	p.d.f.	$-(\log p)^{-1} x^{-1} (1-p)^x, x = 1, 2, \dots$
	m.g.f.	$\log[1 - (1-p)e^t]/\log p, t \in \mathcal{R}$
$L(p)$	Expectation	$-(1-p)/(p \log p)$
	Variance	$-(1-p)[1 + (1-p)/\log p]/(p^2 \log p)$
	Parameter	$p \in (0, 1)$

All p.d.f.'s are w.r.t. counting measure.

Table 1.2. Distributions on $\mathcal{R}$ with Lebesgue p.d.f.'s

Uniform	p.d.f.	$(b-a)^{-1}I_{(a,b)}(x)$
	m.g.f.	$(e^{bt} - e^{at})/[(b-a)t], \quad t \in \mathcal{R}$
$U(a, b)$	Expectation	$(a+b)/2$
	Variance	$(b-a)^2/12$
	Parameter	$a, b \in \mathcal{R}, \quad a < b$
Normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2}$
	m.g.f.	$e^{\mu t + \sigma^2 t^2/2}, \quad t \in \mathcal{R}$
$N(\mu, \sigma^2)$	Expectation	μ
	Variance	σ^2
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$
Exponential	p.d.f.	$\theta^{-1}e^{-(x-a)/\theta}I_{(a,\infty)}(x)$
	m.g.f.	$e^{at}(1-\theta t)^{-1}, \quad t < \theta^{-1}$
$E(a, \theta)$	Expectation	$\theta + a$
	Variance	θ^2
	Parameter	$\theta > 0, \quad a \in \mathcal{R}$
Chi-square	p.d.f.	$\frac{1}{\Gamma(k/2)2^{k/2}}x^{k/2-1}e^{-x/2}I_{(0,\infty)}(x)$
	m.g.f.	$(1-2t)^{-k/2}, \quad t < 1/2$
χ_k^2	Expectation	k
	Variance	$2k$
	Parameter	$k = 1, 2, \dots$
Gamma	p.d.f.	$\frac{1}{\Gamma(\alpha)\gamma^\alpha}x^{\alpha-1}e^{-x/\gamma}I_{(0,\infty)}(x)$
	m.g.f.	$(1-\gamma t)^{-\alpha}, \quad t < \gamma^{-1}$
$\Gamma(\alpha, \gamma)$	Expectation	$\alpha\gamma$
	Variance	$\alpha\gamma^2$
	Parameter	$\gamma > 0, \quad \alpha > 0$
Beta	p.d.f.	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}I_{(0,1)}(x)$
	m.g.f.	No explicit form
$B(\alpha, \beta)$	Expectation	$\alpha/(\alpha+\beta)$
	Variance	$\alpha\beta/[(\alpha+\beta+1)(\alpha+\beta)^2]$
	Parameter	$\alpha > 0, \quad \beta > 0$
Cauchy	p.d.f.	$\frac{1}{\pi\sigma}\left[1+\left(\frac{x-\mu}{\sigma}\right)^2\right]^{-1}$
	ch.f.	$e^{\sqrt{-1}\mu t - \sigma t }$
$C(\mu, \sigma)$	Expectation	Does not exist
	Variance	Does not exist
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$

Table 1.2. (continued)

t-distribution	p.d.f.	$\frac{\Gamma[(n+1)/2]}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{x^2}{n}\right)^{-(n+1)/2}$
	ch.f.	No explicit form
t_n	Expectation	0, ($n > 1$)
	Variance	$n/(n-2)$, ($n > 2$)
	Parameter	$n = 1, 2, \dots$
F-distribution	p.d.f.	$\frac{n^{n/2}m^{m/2}\Gamma[(n+m)/2]x^{n/2-1}}{\Gamma(n/2)\Gamma(m/2)(m+nx)^{(n+m)/2}}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$F_{n,m}$	Expectation	$m/(m-2)$, ($m > 2$)
	Variance	$2m^2(n+m-2)/[n(m-2)^2(m-4)]$, ($m > 4$)
	Parameter	$n = 1, 2, \dots$, $m = 1, 2, \dots$
Log-normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}x^{-1}e^{-(\log x - \mu)^2/2\sigma^2}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$LN(\mu, \sigma^2)$	Expectation	$e^{\mu+\sigma^2/2}$
	Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$
Weibull	p.d.f.	$\frac{\alpha}{\theta}x^{\alpha-1}e^{-x^\alpha/\theta}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$W(\alpha, \theta)$	Expectation	$\theta^{1/\alpha}\Gamma(\alpha^{-1} + 1)$
	Variance	$\theta^{2/\alpha} \left\{ \Gamma(2\alpha^{-1} + 1) - [\Gamma(\alpha^{-1} + 1)]^2 \right\}$
	Parameter	$\theta > 0$, $\alpha > 0$
Double Exponential	p.d.f.	$\frac{1}{2\theta}e^{- x-\mu /\theta}$
	m.g.f.	$e^{\mu t}/(1 - \theta^2 t^2)$, $ t < \theta^{-1}$
$DE(\mu, \theta)$	Expectation	μ
	Variance	$2\theta^2$
	Parameter	$\mu \in \mathcal{R}$, $\theta > 0$
Pareto	p.d.f.	$\theta a^\theta x^{-(\theta+1)}I_{(a,\infty)}(x)$
	ch.f.	No explicit form
$Pa(a, \theta)$	Expectation	$\theta a/(\theta - 1)$, ($\theta > 1$)
	Variance	$\theta a^2/[(\theta - 1)^2(\theta - 2)]$, ($\theta > 2$)
	Parameter	$\theta > 0$, $a > 0$
Logistic	p.d.f.	$\sigma^{-1}e^{-(x-\mu)/\sigma}/[1 + e^{-(x-\mu)/\sigma}]^2$
	m.g.f.	$e^{\mu t}\Gamma(1 + \sigma t)\Gamma(1 - \sigma t)$, $ t < \sigma^{-1}$
$LG(\mu, \sigma)$	Expectation	μ
	Variance	$\sigma^2\pi^2/3$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$